Tiny bits, tens and ones

Electrons

Name _____ Date _____

Show your thinking with numbers, marks, or words.

1. Little sets out 27 electron-model beads, then adds 18 more. How many beads are on the tray?

27 + 18 = _____

Show your thinking.

2. Big has 64 beads for charge models and puts 28 into a bag. How many beads stay out?

64 − 28 = _____

Show your thinking.

3. Little draws 36 electron dots along one model wire and 19 along another. How many dots are drawn altogether?

36 + 19 = _____

Show your thinking.

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Electrons

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4. A craft box has 82 paper bits for the balloon test. Little uses 47. How many bits remain in the box?

82 − 47 = _____

Show your thinking.

5. Big counts 46 small lamp-kit clips in one box and 37 in another. How many clips are there altogether?

46 + 37 = _____

Show your thinking.

6. Little has 95 sock stickers for a static-cling notebook and uses 56. How many stickers are left?

95 − 56 = _____

Show your thinking.

WONDER DAILY

Electrons

Where did the tiny bits go?



Tiny bits. A gentle charge.

GRADE 3

Little Wonderer studied an atom model under the lamp. An atom is a tiny building block of matter. Its nucleus has protons and neutrons packed in the middle. Electrons move around the nucleus. These tiny bits carry electric charge and have much less mass than protons or neutrons.



Big rubbed a balloon on a wool sweater, then held it near paper bits. A few bits lifted. Rubbing can move some electrons from the wool onto the balloon. Electrons carry negative charge. The balloon now had extra electrons; the wool had fewer. The tiny bits had moved, not disappeared.



"Charge can be positive or negative," said Big. Opposite charges pull toward each other. Like charges push apart. Nearby paper can be pulled even when its charges balance overall: the balloon nudges charges inside the paper a little out of place. That makes the nearer side pull toward it.



Next, Big closed a switch on a battery kit. The lamp lit. A wire is a thin metal path. Some electrons in metal can move along it. The closed circuit made a complete loop for electric current. The battery supplied energy; electrons were already in the wires. The lamp did not use them up.



Make a little book.

1. Print one-sided: US Letter, portrait, actual size / 100%. No fit. Headers/footers off.
2. Fold long edge to long edge; reopen. Fold short edge to short edge, then in half again. Open: eight boxes.
3. Cut only the solid center line across the middle two boxes. Keep the outer boxes joined.
4. Refold long edge to long edge, print outside. Push ends together to open a diamond, then a cross.
5. Wrap into a book with page 1 in front. Follow pages 1 to 8.



Look closely, notice the cling, and pause whenever you like.

Your turn

Write one command sentence for the balloon test.

Example: Rub the balloon on your sweater, then hold it near the paper bits.



Little drew the sock and shirt, with an arrow for electrons moving between them. "The tiny bits were already there," Little said. "Some changed places." Big nodded. Little kept one question: "Where do the tiny bits go when a sock stops sticking?"



A sock stayed on a shirt when Little lifted it. "Static cling!" said Big. That means fabrics stick because of electric charge. As some fabrics rub and separate, electrons can move between them. The fabrics can end up with opposite charges. Electrons can also jump across a tiny gap as charge spreads out.

Give the balloon test its words

Electrons · Curiosity day

Name _____ Date _____

Say one clear command, then draw or write it.



Too small to see



A few electrons move



A sock that stays

Draw a balloon near paper bits. Add an arrow to show what you notice.

rub · hold · near · balloon · paper · electron · charge · cling

First, … tell how to set out the model beads

Then, … tell how to hold the balloon near the paper

Look for, … tell what to notice about a clinging sock

Does your command tell what to do, using a word such as Rub, Hold, or Notice?